

TSPI — Open Questions & Blockers

TSPI AI Engine — Open Questions, Blockers & Feedback on the Latest Files

Prepared for: the TSPI domain-expert team **Status:** 1 hard blocker · 1 data blocker · 12 open questions
Purpose: everything still needed from the clinical team, plus our feedback on the three files received 15 Jul 2026.

1. Summary — where we are

Your Round-3 response resolved the **critical axis-dictionary conflict**, and we have already re-based the engine on the new master. Five phases of engine work are complete and tested (54/54 passing). **The engine is now blocked on data, not on decisions.**

Count		
✓	13 of 13 Round-3 questions	answered and implemented
◆	1 hard blocker	the updated module file still uses legacy axis numbering
◆	1 data blocker	the 180-Network master is not machine-readable
!	12 open questions	mostly master datasets + confirmations

2. Documents received (all read in full)

Ref	File	Pages	Date
[D]	tspi developer answer.pdf	3	9 Jul 2026
[C]	TSPI AI Engine – Domain Expert Answers for the Dev_260713_095402.pdf	22	13 Jul 2026, 09:54
[A]	tspi comment for phase3 developer _260713_104231.pdf	36	13 Jul 2026, 10:42
[B]	tspi comment for phase3 developer .pdf	55	13 Jul 2026, 11:17
[E]	TSPI 3 – Complete Domain Expert Response for the Development Team.pdf	30	15 Jul 2026
[F]	last 39 [Thai: 39 axes, 12 domains, 3 keys] (1)_260715_110654.pdf	81	15 Jul 2026
[G]	200 module tspi 2 (10-7-69).th.en(1).th.en(4) [Thai: revised].xlsx	966 rows	15 Jul 2026

3. Feedback on the three latest files

[E] Round-3 Complete Response — ✓ excellent, no issues

All 13 questions answered, and the **Final Developer Lock** gave us exactly what we needed to proceed. Particularly valuable corrections we would not have arrived at ourselves:

- **Safety is a GATE, not a score** — we would have wrongly subtracted 10 points instead of excluding.
- **Data Completeness must be weighted** (CORE=3 / SUPPORTING=2 / OPTIONAL=1) — you explicitly rejected our `present ÷ expected` assumption. Correct: a missing CORE marker must not look "nearly complete".
- **Evidence splits into two fields** (Status and Grade) — we had conflated them.

- **Module ceiling ~6** — this resolved our 64-module output. "Matching is not prescribing" is now enforced in code.
- **Learning must be propose-only** — our Phase-4 loop was auto-updating the global model from single patients. That was a genuine compliance violation; it is now fixed.

All of the above are implemented. Nothing in [E] is disputed.

[F] 39 Axes / 12 Domains / 3 Keys master — ✓ canonical, 2 observations

This is now our **frozen ontology** (`framework_version: 39-axis-master-260715`). The "Key Correction" on p.41 independently confirms [E] — the two agree, which resolved the conflict.

Two things to be aware of:

1. **The file contains two different axis lists.** Pages **36-41** carry the corrected list (A35 = Visceral Organ Functional Reserve, A37 = Oncologic Cell Fate...). Pages **42-60** carry an **older detailed section** (A35 = "Organ Resilience & Functional Reserve"; A36-A39 absent) — *this older section is the one that contains the per-axis `Module:` lists. We used pp.36-41 only.* → see **Q13**.
2. **The 180-Network table is in there** (pp.61-81) — genuinely good news, and we confirmed **CELA = network #120**. But it is a visual Thai/English prose table: our best automated extraction recovered only **102 of 180** rows with mixed-up columns, so we did **not** load it. → see **Blocker 2**.

[G] Updated 200-module xlsx — ♦ this is the blocker

[E] Q2 states: "The updated Official Module Registry will directly contain the correct current Axis codes for every module." **This file does not.** Verified programmatically across all 242 product codes:

Check	Result
Rows using legacy axis numbering	54
Rows using new axis numbering	29
Rows mixing both	2
Axis numbers whose name disagrees with the master	20 of 39
Rows still containing deprecated " Protein Quality Control "	21
Rows still containing deprecated " Protein Clearance / Proteotoxic Stress "	22
Rows still containing deprecated " Cognitive-Emotional Loop " as an axis	14

Your own example, H-001 Curcuma, still reads: Axis 16 – Liver Detoxification · Axis 17 – Digestive Function · Axis 18 – Microbiome Ecology · Axis 19 – Gut Barrier · Axis 21 – Endothelial Function · Axis 25 – Tissue Repair · Axis 27 – Metabolic Balance Under the master these should be **A15 · A16 · A17 · A18 · A20 · A24 · A27** (and A27 changes meaning entirely — legacy "Metabolic Balance" vs current "Glycation / Carbonyl Stress").

H-006 BENCHAKUL still contains AXIS 37 Protein Quality Control System , AXIS 38 Protein Clearance & Proteotoxic Stress , Axis 39 Cognitive-Emotional Loop — the three definitions [E] just deprecated.

Per your own rule, this file **cannot enter production**, so we have quarantined our module→axis map (`production_allowed: false`). Module recommendations are therefore **not clinically usable** today.

4. ♦ BLOCKERS

Blocker 1 — The module registry still carries legacy axis numbering

What we need: an Official Module Registry where `target_axes` already contains **current** A1–A39 codes, with `mapping_status = APPROVED`.

Why we cannot fix it ourselves: [E] Q2 is explicit — *"Do not use a blind numerical conversion as the Production source of truth... Each legacy mapping must be reviewed semantically."* A numeric shift would silently produce clinically wrong mappings (A27 is the proof: same number, different meaning).

Question: is a further-corrected registry coming — or would you like us to produce a **candidate remap** (legacy → current, with `conversion_type` and confidence flags) for your team to review and sign off? We can generate that within a day; it would still need clinical approval before use.

Easiest format: `module_code(H-Code)`, `module_name`, `phytocore_code`, `target_axes[]` (current codes), `dose_type`, `contraindications`, `evidence_grade`, `status`, `axis_master_version`.

Blocker 2 — The 180-Network master is not machine-readable

What we need: the 180 Living Biological Networks as **CSV/Excel** (the table already exists in [F] pp.61–81 — we simply cannot extract it reliably).

Why it matters: [E] Q3 makes the network layer **mandatory architecture**, and Part 6 of [B] states TSPI must reason *from networks toward symptoms*. Our Module Match v1.0 allocates **25% to network match** — until the master lands, **every module scores 0 on that component**.

Easiest format (your own schema from [E] Q3): `network_code`, `network_number`, `network_name_en`, `network_name_th`, `primary_axis_code`, `secondary_axis_codes[]`, `domain_code`, `network_type`, `upstream_network_codes[]`, `downstream_network_codes[]`, `accepted_symptoms[]`, `accepted_markers[]`, `evidence_grade`, `status`.

5. Master datasets still required

(You listed these yourself in [E] Part 3. We have built the schemas + loaders; we need the clinical content.)

#	Dataset	Blocks	Our status
Q1	Marker Registry — marker → primary/secondary axis + direction rule (5 types)	generic lab handling	~12 markers seeded by us
Q2	Laboratory Cut-off Registry — optimal/subclinical/functional/pathological + critical values	accurate axis severity	thresholds are placeholders
Q3	Symptom→Axis / Symptom→Network Dictionary	the phenotype engine	16 symptoms seeded from your examples — needs ratification
Q4	Red-Flag Registry	patient safety screening	24 symptom + 3 condition + 10 critical-value rules seeded from your examples — needs ratification
Q5	Evidence Grade Registry (A–D per module + claim + target)	module ranking	no grades exist in the registry — every module currently defaults to D (weakest)
Q6	NSS Coefficients after clinical validation	final scoring	v0.1 implemented behind a version flag

6. Clarifications & confirmations

Q7 — What are the official sub-axis codes for A10? [E] lists A10A–A10D (Protein Folding / ER Stress–UPR / Autophagy–Lysosome / Proteotoxic Burden) and [F] shows 4 sub-axes for A10. Please confirm the exact **codes** so we key them identically.

Q8 — What is `minimum_evidence` for each axis? [E] Q9 says this must differ per axis but does not enumerate it. Will it come with the Axis Master, or shall we propose defaults for your review?

Q9 — Treatment-step compatibility (5% of Module Match). We need the mapping of **modules** → **the 9 Restoration Steps** to score this. Does it exist? Until then this component is neutral for every module.

Q10 — Historical response (5% of Module Match). This requires past outcome data per module. Should it stay neutral until enough outcomes accumulate, or do you have historical response data we can load?

Q11 — Population-learning minimum sample size. [E] Q11 requires a "minimum sample size" but does not give a number. **We used 30** as a placeholder. Please confirm or replace.

Q12 — Domain→Key relationship weights. [E] Q6 gives primary/secondary keys per domain and says weights "*may remain descriptive until clinically validated*". We have stored them as descriptive only — please confirm that is right.

Q13 — The older axis section inside [F] (pp.42–60). That section uses the **old** A35 name and omits A36–A39 — but it is also the section carrying the per-axis `Module:` lists. **We ignored it entirely.** Please confirm it is superseded, and tell us whether those per-axis module lists should be used (re-mapped) or discarded.

Q14 — Interim handling of the 25% network-match weight. Until the Network master arrives, every module scores 0 on network match, so all scores sit ~25% lower (ranking is unaffected, since it is uniform). Prefer we (a) leave it, or (b) temporarily re-normalise the remaining weights to sum to 1.0? We recommend **(a) leave it** — it keeps the approved formula intact and honestly shows the data gap.

7. What we built in the meantime — please ratify

To keep moving, we seeded provisional datasets **verbatim from your own worked examples**. Each is labelled `authoritative: false` and is excluded from clinical authority. **These need your review:**

File	Contents	Source
<code>red_flag_rules.json</code>	24 symptom rules · 3 condition rules · 10 critical-value rules	your Q5 examples + section 10 of [A]/[B]
<code>symptom_dictionary.json</code>	16 symptoms · 3 clusters · weights 0.25/0.50/0.75/1.00	your Q4 examples (bloating, dizziness, fatigue)
<code>negative_test_rules.json</code>	3 rules (gastroscopy, brain MRI, routine bloods)	your section 8 examples
<code>nss_config.json</code>	NSS v0.1 coefficients + NetworkFactor bands	your Q3 formula
<code>module_match_config.json</code>	the approved 30/25/15/10/10/5/5 weights + ceiling	your Q10

Specific ratification requests:

- Are the **critical-value thresholds** acceptable as an interim? ($K^+ <2.5 / >6.5$ · $Na^+ <120 / >160$ · glucose $<54 / >450$ mg/dL · $Hb <7$ · platelets <20 · ALT/AST >1000 · $SpO_2 <88\%$ · troponin >0.04)
- Is the **symptom weighting** faithful to your intent? (*bloating alone* = *nonspecific*; + *early satiety* + *belching* → A16 high; + *constipation* → A17/A19 rise; *worse with stress* → A34 + CELA)

8. What your answers unlock

Input	Unlocks
Blocker 1 — corrected module registry	clinically usable module recommendations (currently quarantined)
Blocker 2 — 180-Network CSV	the mandatory network reasoning layer + the 25% network-match component
Q1/Q2 Marker + cut-offs	accurate per-axis severity from any lab
Q3/Q4 Symptom dictionary + red flags	ratified clinical content behind the phenotype and safety engines
Q5 Evidence grades	correct module ranking (all modules are currently graded D)
Q7-Q14	final structural details + confirmation of our interim choices

How to send: CSV/Excel is ideal for the registries; a short note is perfect for Q7-Q14. We load them directly — **most require no software changes**, because every dataset is versioned and loaded from file.

9. For reference — what is already implemented from your feedback

Deterministic engine (**the LLM never assigns an axis score, network identity or module name**) · official-registry-only module resolution · physician approval gate · de-identification + audit · **no default-50** · explicit `NOT_ASSESSED` (no data is never "normal") · the four dimensions (Evidence Status, Evidence Grade, Confidence, Data Completeness) + **Biological Uncertainty** · HYPOTHESIS never scores · **red-flag screening runs first** · critical alerts survive a low NSS · **NSS v0.1** (confidence-weighted, NetworkFactor, no universal inflammation multiplier) · **Module Match v1.0** (safety gate, mechanism de-dup, ~6 ceiling, `reason_not_selected`) · **symptoms as first-class evidence** (a symptoms-only patient is now assessable) · **propose-only population learning** with clinician approval · everything version-stamped.